8 **(a)** A loudspeaker, microphone and datalogger are arranged as shown in Fig. 8.1.



Fig. 8.1

The loudspeaker is emitting a sound wave which is detected by the microphone connected to the datalogger. The variation with time t of the voltage V of the signal captured by the datalogger as shown in Fig. 8.2.

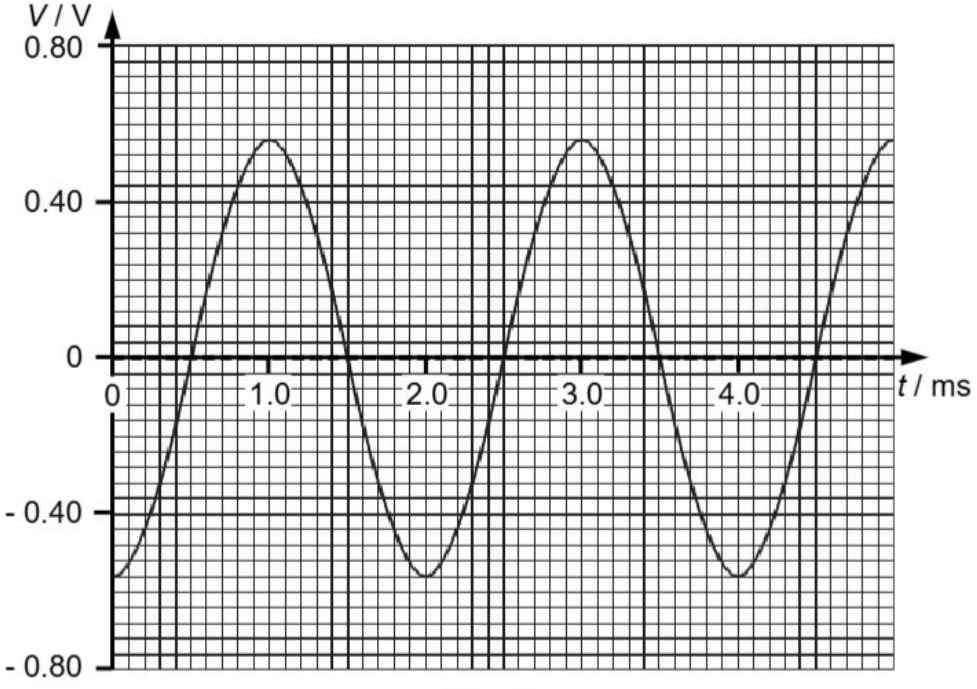


Fig. 8.2

(i) Calculate the frequency of the sound wave.

frequency = ... Hz [1]

(ii) The intensity of the sound wave is reduced to a quarter of its original intensity without a change in frequency. Assume that the amplitude of the signal received by the datalogger is proportional to the amplitude of the sound wave.

On Fig. 8.2, sketch the trace that is now seen on the screen of the datalogger. [2]

(b) A metal sheet is now placed in front of the loudspeaker in **(a)**, as shown in Fig. 8.3.

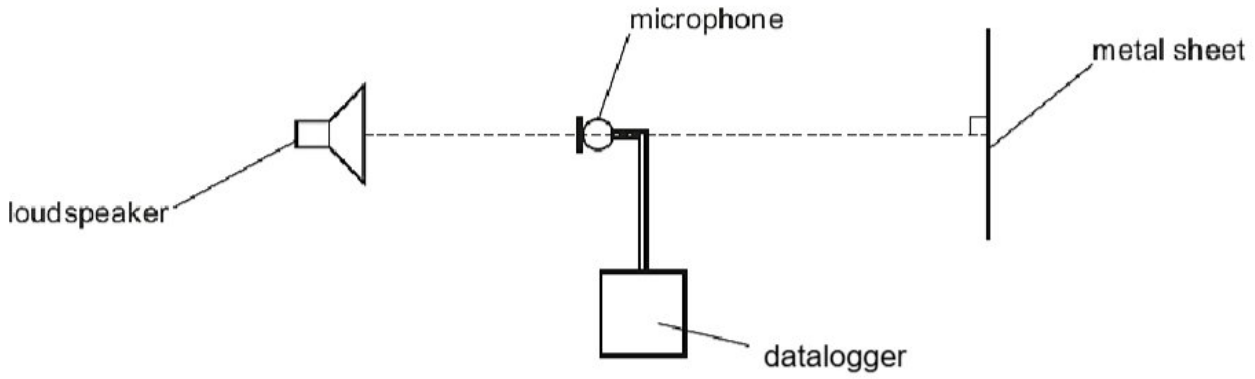


Fig. 8.3

A stationary wave is formed between the loudspeaker and the metal sheet.

The initial position of the microphone is such that the trace on the datalogger has an amplitude minimum. It is now moved a distance of 1.05 m from its initial position along the line joining the loudspeaker and metal sheet.

As the microphone moves, it passes through three positions where the trace has an amplitude maximum before ending at a position where the trace has an amplitude minimum.

(i) Explain how the stationary wave is formed from the incident sound wave.

.....

.....

.....

..... [2]

(ii) Determine the speed of the sound in the air.

speed = ... m s^{-1} [3]

(iii) The metal sheet in Fig. 8.3 is replaced with a material that absorbs some of the sound wave. State and explain what happens to the amplitude registered at the minimum position on the datalogger.

.....

.....

.....

..... [2]

(c) An arrangement for demonstrating the interference of light is shown in Fig. 8.4.

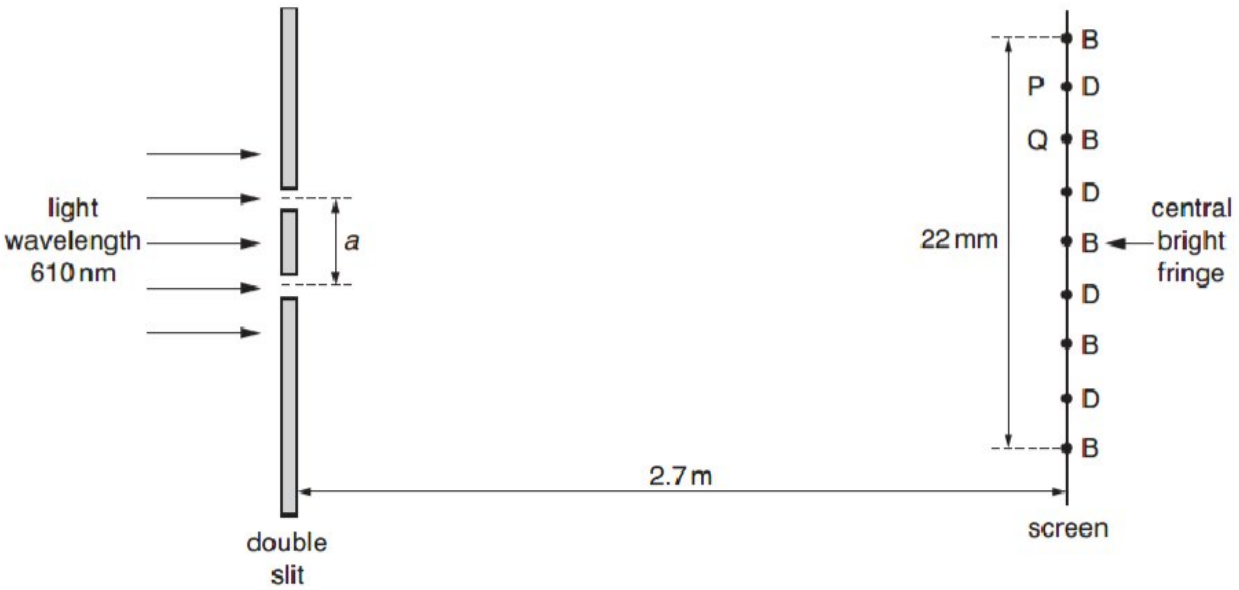


Fig. 8.4

The wavelength of the light is 610 nm. The distance between the double slit and the screen is 2.7 m.

An interference pattern of bright fringes and dark fringes is observed on the screen. The centres of the bright fringes are labelled B and centres of the dark fringes are labelled D. Point P is the centre of a particular dark fringe and point Q is the centre of a particular bright fringe, as shown in Fig. 8.4. The distance across five bright fringes is 22 mm.

(i) **1.** State the phase difference between the waves meeting at Q.

phase difference = ... ° [1]

2. Calculate the path difference, in nm, of the waves meeting at P.

path difference = ... nm [1]

(ii) Determine the distance *a* between the two slits.

a = ... m [2]

(iii) The monochromatic light source of wavelength 610 nm is now replaced with a white light source.

Describe the appearance of the central fringe and the fringes observed on either side of the central fringe.

.....

.....

.....

..... [2]

(d) A diffraction grating is used with different wavelength of visible light. The angle θ of the fourth-order maximum from the zero-order central maximum is measured for each wavelength. The variation with wavelength λ of sin θ is shown in Fig. 8.5.

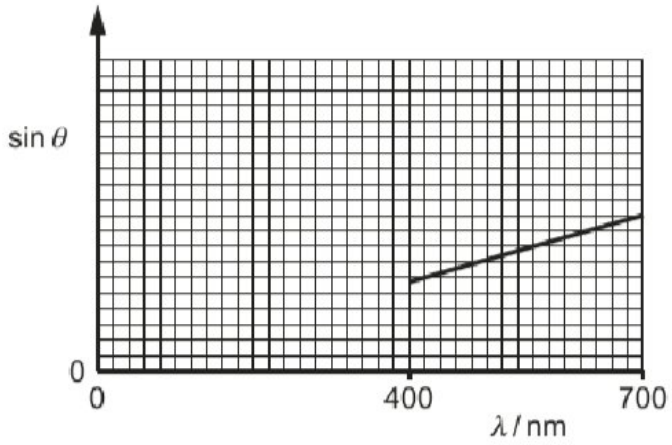


Fig. 8.5

(i) The gradient of the graph is G.

Determine an expression, terms of G, for the distance d between the centres of two adjacent slits in the diffraction grating.

d = ... [2]

(ii) On Fig. 8.5, sketch a graph to show the results that would be obtained for second-order maxima. [2]

[Total: 20]